

Erdős Problem #853

EP-853

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $d_n = p_{n+1} - p_n$, where p_n is the n th prime. Let $r(x)$ be the smallest even integer t such that $d_n = t$ has no solutions for $n \leq x$.

Is it true that $r(x) \rightarrow \infty$? Or even $r(x)/\log x \rightarrow \infty$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>